

9 Review

9.3 Review of Finance

For these formulas, we have the following: P_0 is the principal. r is the interest rate. t is the number of time intervals (specified by problem) which have passed. A is the amount paid back. I is the interest.

Simple Interest	$A = P_0(1 + r)$
Simple Interest over Time	$A = P_0(1 + rt)$

For these formulas, we have the following: P_0 is the principal, or amount of a loan. r is the annual interest rate. t is the number of years which have passed. A is the total amount. k is the number of times interest is compounded per year. d is the amount we deposit (or withdraw) from the account regularly, or the payment amount on a loan.

Compound Interest	$A = P_0 \left(1 + \frac{r}{k}\right)^{kt}$
Savings Annuity	$A = d \frac{\left(1 + \frac{r}{k}\right)^{kt} - 1}{\frac{r}{k}}$
Payout Annuity/Loans	$P_0 = d \frac{1 - \left(1 + \frac{r}{k}\right)^{-kt}}{\frac{r}{k}}$

Example 9.3.1 Marcy received an inheritance of \$20,000, and invested it at 4% interest. She is going to use it for college, withdrawing money for tuition and expenses each quarter. How much can she take out each quarter if she has 4 years of school left?

Example 9.3.2 Paul wants to buy a new car. Rather than take out a loan, he decides to save \$250 a month in an account earning 2% interest compounded monthly. How much will he have saved up after 4 years?

Example 9.3.3 Keisha is managing investments for a non-profit company. They want to invest some money in an account earning 3% interest compounded once every 6 months with the goal to have \$30,000 in the account in 10 years. How much should Keisha deposit into the account?

Example 9.3.4 Miao is going to take out a loan for some new office equipment at a 3% rate over a 6 year term. If she can afford monthly payments of \$100, how much new equipment can she buy?

Example 9.3.5 How much would you need to save every quarter in an account earning 4% interest to have \$5,000 saved up in 4 years?